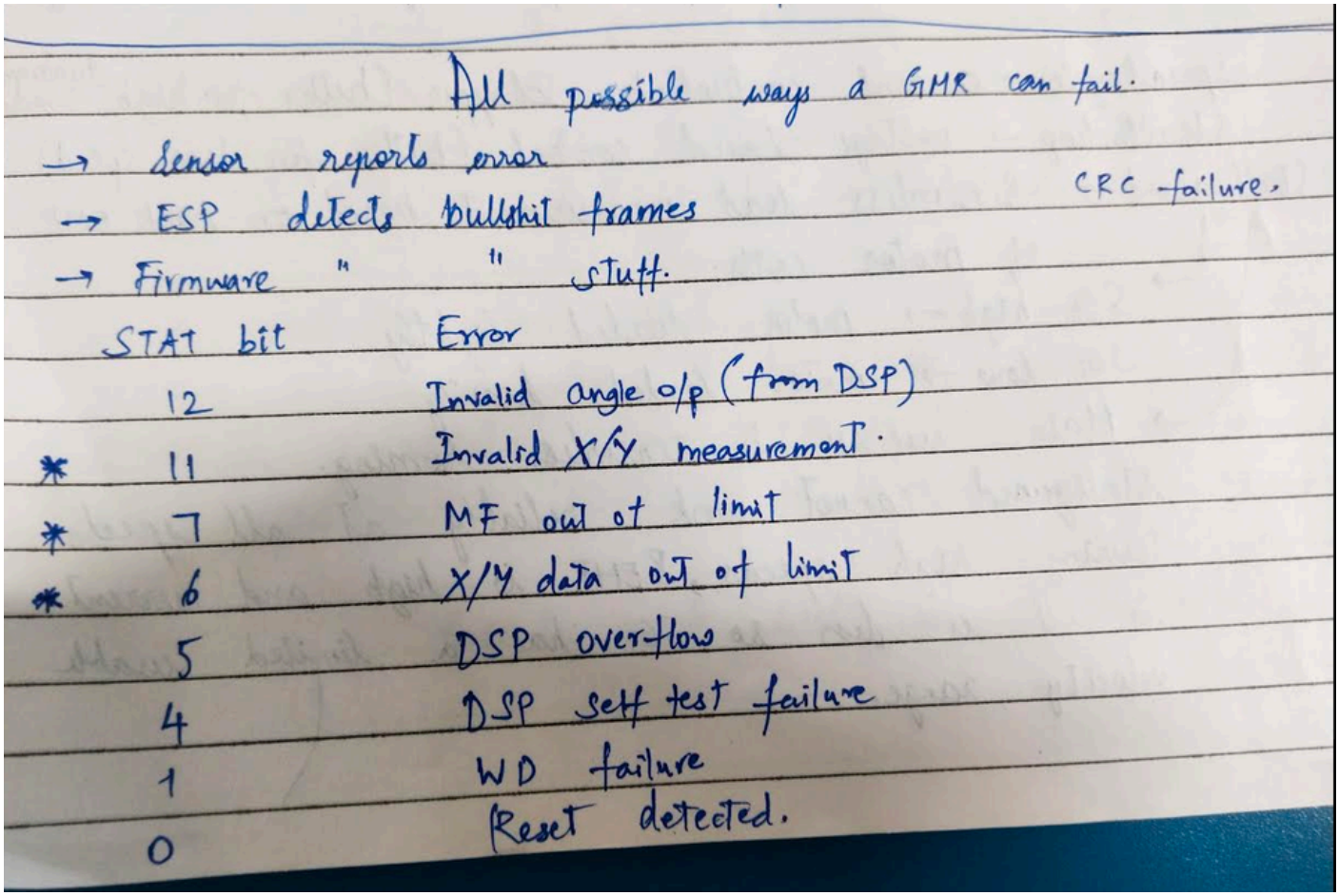
This document is solely written wrt GMR Sensor ONLY.

2.2.6 Safety features

The TLE5012B offers a multiplicity of safety features to support the Safety Integrity Level (SIL) and
it is a PRO-SIL™ product.
Safety features are:
• Test vectors switchable to ADC input (activated via SSC interface)
• Inversion or combination of filter input streams (activated via SSC interface)
• Data transmission check via 8-bit Cyclic Redundancy Check (CRC) for SSC communication and 4-bit CRC
nibble for SPC interface
• Built-in Self-test (BIST) routines for ISM, CORDIC, CCU, ADCs run at startup
• Two independent active interfaces possible
• Overvoltage and undervoltage detection
Disclaimer
PRO-SIL™ is a Registered Trademark of Infineon Technologies AG.
The PRO-SIL™ Trademark designates Infineon products which contain SIL Supporting Features.
SIL Supporting Features are intended to support the overall System Design to reach the desired SIL (according
to IEC61508) or A-SIL (according to ISO26262) level for the Safety System with high efficiency.
Data Sheet 7 Rev. 2.1
2018-06-20
TLE5012B
GMR-Based Angle Sensor
Functional description
SIL respectively A-SIL certification for such a System has to be reached on system level by the System
Responsible at an accredited Certification Authority.
SIL stands for Safety Integrity Level (according to IEC 61508)
A-SIL stands for Automotive-Safety Integrity Level (according to ISO 26262)
Above is fully fetched from the datasheet



You get error information from the TLE5012B in three layers over SSC, and a fourth you build yourself from other registers

Read AVAL register during normal operation, and STAT register for identifying sensor faults.

STAT flag	What it tells you
`S_RST`	A reset event occurred.
`S_WD`	The watchdog expired.
`S_VR`	A monitored voltage/ground-related fault was detected.
`S_FUSE`	A fuse/configuration integrity check failed.
`S_DSPU`	A DSP-related self-test failed or is still running.
`S_OV`	DSP overflow occurred.
`S_XYOL`	X/Y measurements exceeded their specified limits.
`S_MAGOL`	Magnetic vector magnitude is outside its limits.
`S_ADCT`	ADC diagnostic test failed or requires investigation.
`S_ROM`	ROM integrity check failed or is still running.
`NO_GMR_XY`	Valid X/Y measurements are unavailable.
`NO_GMR_A`	A valid angle is unavailable.

See particular register addresses in the datasheet of GMR Sensor.

There are additional registers for detailed diagnostics

Register	Address	Purpose	Related errors
ACSTAT	`0x01`	Check which diagnostic and self-test functions are activated.	Disabled diagnostics; startup/configuration checks
D_MAG	`0x14`	Read the magnetic vector magnitude.	Magnet too weak/strong; `ERR_MAGNET_MARGIN`
ADC_X	`0x10`	Read raw X-axis GMR measurement.	Abnormal X measurement; investigate `S_XYOL`
ADC_Y	`0x11`	Read raw Y-axis GMR measurement.	Abnormal Y measurement; investigate `S_XYOL`
T_RAW	`0x15`	Read the raw internal temperature measurement.	`ERR_TEMPERATURE_LIMIT`
MOD_1	`0x06`	Check interface configuration.	`ERR_CONFIG_MISMATCH`
MOD_2	`0x08`	Check configuration, including angular direction and other operating settings.	`ERR_CONFIG_MISMATCH`
MOD_3	`0x09`	Check additional sensor configuration.	`ERR_CONFIG_MISMATCH`